

Ziye Di

Direct Ph.D. Candidate

School of Integrated Circuits and Micro-nanoelectronics, Fudan University

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Profile

Ziye Di is a Direct Ph.D. candidate in Electronic Science and Technology at Fudan University, School of Integrated Circuits and Micro-nanoelectronics.

Research interests include layered ferroelectrics, integrated circuits, in-memory computing, neuromorphic hardware, and bio-inspired electronic systems.

Education

2022.09 - 2027.06 | Fudan University | Direct Ph.D. in Electronic Science and Technology | School of Integrated Circuits and Micro-nanoelectronics

2018.09 - 2022.06 | Central South University | B.S. in Microelectronic Science and Engineering

Core courses: Microelectronic Materials (A-), Principles of Integrated Chip Manufacturing Process (A), Principles of Ferroelectric Device Design (A), Semiconductor Memory Materials, Devices and Processes (A).

Project Experience

Polarization-Dependent Conductivity Mechanism in Layered Ferroelectrics and In-Memory Computing Applications: published one Nature Nanotechnology paper as first author. Combined theoretical calculations and experiments to reveal the microscopic mechanism of asymmetric conducting paths generated by the built-in electric field, also known as the hidden Stark effect. Modulated transistor conductivity thresholds and realized electrically self-switching logic-in-memory functions including AND, OR, XOR, and NAND.

Designed and fabricated dual-gate layered ferroelectric channel field-effect transistors (FeCFETs) with an asymmetric equivalent oxide thickness dielectric architecture. Characterized transfer/output curves, threshold voltage shifts, hysteresis memory windows, and high/low resistance states, then constructed a complex logic network based on 2T1R cells.

Electronic Spinal Cord System Based on Neuromorphic Primitive Circuits and Motor Function Reconstruction: first-author Nature Nanotechnology manuscript under revision. Established an ultra-low-power brain-circuit-muscle closed loop, enabling rats to restore volition-driven forward and backward voluntary motor functions without external supervision within 14 days.

Constructed a Neuromorphic Primitive Circuit (NPC) integrating ferroelectric floating-gate transistors and threshold switching memristors. Led device design and in vivo animal experiments, and used MATLAB for bioelectrical signal processing and Local Field Potential analysis.

Academic Publications

1. Ziye Di, Jinquan Ma, Bo Hu, Xiaolong Ma, Keyi Chen, Sifan Chen, Bicheng Wang, Yang Wang, Chunsen Liu, Huajiang Chen*, Shuiyuan Wang*, Peng Zhou*. Neuromorphic Primitive Circuit for Restoration of Volitional-driven Voluntary Locomotion. Nature Nanotechnology, under revision. SCI Q1 TOP, IF=38.3.
2. Ziye Di, Ruge Quhe, Jiaxin Zhang, Yuxuan Sun, Lingxue Zhang, Ying Guo, Shuiyuan Wang*, Peng Zhou*. Asymmetric conducting route and potential redistribution determine the polarization-dependent conductivity in layered ferroelectrics. Nature Nanotechnology, 2023.11. SCI Q1 TOP, IF=38.3.
3. Ziye Di, Siwei Xue, Shuiyuan Wang*, Tianxiang Wu, Nuo Xu, Yibo Sun, Chaofan Zeng, Shunli Ma*, Peng Zhou*. Hybrid neuromorphic hardware with sparing 2D synapse and CMOS neuron for character recognition. Science Bulletin, 2023.10. SCI Q1 TOP, IF=18.9.

4. Ziye Di, Zifei Gao, Xiaofan Zhang, Shuiyuan Wang*, Peng Zhou*. Emerging neuromorphic devices and circuits for bio-inspired electronics. Journal of Semiconductors, 2026.4. IF=5.3.
5. Xiaoxian Liu, Shuiyuan Wang, Ziye Di, Haoqi Wu, Chunsen Liu, Peng Zhou*. An Optoelectronic Synapse Based on Two-Dimensional Violet Phosphorus Heterostructure. Advanced Science, 2023.05. SCI Q1 TOP, IF=14.3.
6. Xiaolong Ma, Ziye Di, Xianghong Zhang, Shuiyuan Wang. A scalable neuromorphic platform with four neural motifs featuring <1 ms Per-Neuron Latency. 2026 IEEE Electron Devices Technology and Manufacturing Conference (EDTM), 2026. International Conference.

Conference and Academic Exchanges

Attended the 10th IEEE Electron Devices Technology and Manufacturing Conference (EDTM 2026) and gave an oral presentation. March 2026, Malaysia.

Attended the Young Elite Scientists Sponsorship Program "Leading Action" Science and Technology Symposium, served as a group leader, and gave an oral presentation. January 2026, Beijing.

Attended The 10th International Conference on Technological Advances of Thin Films & Surface Coatings (Thin Films 2024), participating in thematic reports and academic exchanges. July 2024, Singapore.

Professional Skills

Complete CMOS device fabrication process, including EBL, magnetron sputtering, and RIE-ICP etching.

Semiconductor device parameter analyzers such as Keysight B1500A for electrical performance and electrophysiological testing.

TCAD and SPICE simulation tools. Data processing and layout tools: Origin, Python, C++, MATLAB, L-Edit, and CleWin.

Honors and Awards

2024 | Young Elite Scientists Sponsorship Program by CAST

2024 | National Scholarship for Doctoral Students

2022-2026 | Outstanding Scholarship of Fudan University

2022-2025 | Outstanding Student of Fudan University

Class of 2022 | Outstanding Graduate of Central South University

Leadership and Extracurricular Activities

Participated in the Academic Career Department of the School of Microelectronics and hosted the Doctoral Student Forum during the 2023-2024 academic year.